SCIE2500 Final Report: Curried Functional Spaces

Roman Maksimovich, ID: 21098878
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Supervisor: JIN, Tianling

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1. Introduction

There are essentially two ways to represent a multivariable function. One is to complicate the domain, in which case we write $f : A \times B \to C$, and the other is to complicate the codomain by writing $g : A \to C^B$, such that $g(a)$ is again a function that maps B to C. In programming, the latter approach is called *partial application* or *currying*, and we will adopt this terminology here.

Classical multivariable calculus is built on the former, "uncurried" approach to defining functions of multiple arguments, and in this paper we build the theory of real multivariable calculus based on curried function spaces.

We will be using the topological language of *filters* throughout the work, to simplify proofs. For the definition and properties of filters, see Appendix A.

2. Definitions of curried spaces

2.1. General functions

First, we introduce the most inclusive spaces of curried functions, endowed with the topology of pointwise convergence.

Definition 2.1.1. (pointwise convergence topology, see[1], page 277) Let X be a set, Y a topological space, and denote by $\mathcal{S}(X, Y)$ the set of functions from X to Y. For each $x \in X$ and each open set $U \subset Y$, let $V(x, U)$ be the set of all functions $f \in \mathcal{S}(X, Y)$ such that $f(x) \in U$. The collection

$$\{V(x, U) \mid x \in X, U \subset Y\}$$

forms a subbase of a topology on $\mathcal{S}(X, Y)$, called the *topology of pointwise convergence.* A filter $F \in \mathcal{F}(\mathcal{S}(X, Y))$ will converge to a function $f \in \mathcal{S}(X, Y)$ if and only if $F(x)$ converges to $f(x) \in Y$ for all $x \in X$.

Proposition 2.1.2. (see[1], page 281) *Let X be a set and Y a Hausdorff topological space. Then $\mathcal{S}(X, Y)$ is also Hausdorff.*

Definition 2.1.3. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The set $H_n(\mathbb{R})$ of *curried n-variable functions* is a topological vector space defined recursively as follows:

- If $n = 0$, we set $H_0(\mathbb{R}) = \mathbb{R}$;

- If $H_n(\mathbb{R})$ is defined, we let $H_{n+1}(\mathbb{R})$ to be the set of all functions from $\mathbb{R}$ to $H_n(\mathbb{R})$. This set is given the pointwise linear structure and the topology of pointwise convergence, such that a filter $F \in \mathcal{F}(H_{n+1}(\mathbb{R}))$ converges to a function $f \in H_{n+1}(\mathbb{R})$ if and only if $F(x) \to f(x) \in H_n(\mathbb{R})$ for all $x \in \mathbb{R}$. Proposition 2.1.2 and Lemma B.1 ensure that $H_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space.

2.2. Continuous functions

Next, we define the spaces $C_n(\mathbb{R})$ of continuous curried functions. Since the continuity of a function depends on the topology of the codomain, and since curried spaces are defined inductively, we see that the choice of *topology* on $C_n(\mathbb{R})$ influences which functions are included in $C_{n+1}(\mathbb{R})$. A natural choice is the *compact-open topology,* since, if $\mathcal{C}(X, Y)$ is the space of continuous functions between X and Y endowed with this topology, one has

$$\mathcal{C}(X \times Y, Z) \cong \mathcal{C}(X, \mathcal{C}(Y, Z)).$$

Definition 2.2.1. (compact-open topology, see[1], page 301) Let X and Y be two topological spaces, and by $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y. For each compact set $K \subset X$ and each open set $U \subset Y$, let $V(K, U)$ be the set of all functions $f \in \mathcal{C}(X, Y)$ such that $f(K) \subset U$. The collection

$$\{V(K, U) \mid K \subset X, U \subset Y\}$$

forms a subbase of a topology on $\mathcal{C}(X, Y)$, called the *compact-open* topology.

Remark 2.2.2. Clearly, the topology of pointwise convergence on $\mathcal{C}(X, Y)$ is coarser than the compact-open topology.

Corollary 2.2.3. *Let X and Y be topological spaces such that Y is Hausdorff. Then $\mathcal{C}(X, Y)$ is Hausdorff.*

We also have an equivalent definition of the compact-open topology in terms of filters:

Proposition 2.2.4. (see[2], page 34) *Let X and Y be topological spaces such that X is locally compact and Y is regular. Then a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$ converges to a function $f \in \mathcal{C}(X, Y)$ if and only if, for any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have $F(P) \to f(p)$ in Y, where the filter $F(P)$ is based on*

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

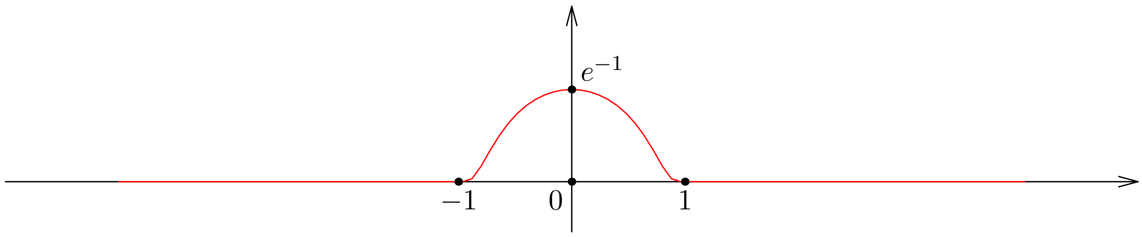
Finally, we give the definition of continuous functions:

Definition 2.2.5. Let $n \in \mathbb{N}_0$. The set $C_n(\mathbb{R})$ of *curried continuous n-variable functions* is a Hausdorff topological vector space defined recursively as follows:
- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$;
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$, endowed with pointwise linear structure and the compact-open topology. Corollary 2.2.3 and Lemma B.2 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space.

Remark 2.2.6. The topology on $C_n(\mathbb{R})$ is finer than the topology induced from $H_n(\mathbb{R})$, and does not coincide with it, unless $n = 0$. To see this, take $n = 1$ and consider the classical function

$$\varphi(x) = \begin{cases} \exp\left(\frac{1}{|x|^2 - 1}\right), & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

This is a $C^\infty(\mathbb{R})$ function supported on $[-1, 1]$. Its graph looks as follows:



Now consider the following sequence of functions:

$$f_k(x) = e \cdot \varphi\left(k \cdot \left(x - \frac{1}{k}\right)\right).$$

Clearly, f_k is infinitely differentiable on $\mathbb{R}$, supported on $[0, 2/k]$, and has a maximum at $x = 1/k$ with $f_k(1/k) = 1$. It is clear that for all $x \in \mathbb{R}$, we have $f_k(x) \xrightarrow[k\to\infty]{} 0$, so $f_k \to 0$ in the topology induced from $H_1(\mathbb{R})$. However, f_k do not converge to 0 in the native topology of $C_1(\mathbb{R})$. To see this, take $x_k = 1/k$ and observe that $f_k(x_k) \equiv 1 \neq 0$.

A similar argument applies for any other $n \geq 2$. Hence we conclude that $C_n(\mathbb{R})$ is not a subspace of $H_n(\mathbb{R})$.

Lemma 2.2.7. *Let X, Y be topological spaces such that X is locally compact. Then a sequence $\{f_k\}_{k\in\mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ if and only if $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x$ in X.*

Proof: First, assume that $f_k \to f \in \mathcal{C}(X, Y)$, and consider a sequence $\{x_k\}_{k=1}^{\infty} \subset X$ converging to $x \in X$. Let U be a neighborhood of $f(x)$ in Y. Since X is locally compact, the point $x \in X$ has a compact neighborhood K. Hence, eventually as $k \to \infty$, we have $x_k \in K$ and $f_k \in V(K, U)$, which means that eventually $f_k(x_k) \in U$. Since U was arbitrary, we conclude that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$.

Conversely, suppose that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x \in X$. To prove that $f_k \to f \in \mathcal{C}(X, Y)$, consider a neighborhood $V(K, U)$ of f, and assume that there is a subsequence $\left\{f_{k_l}\right\}_{l=1}^{\infty}$ such that $f_{k_l}(K) \not\subset U$. That means that we can find a sequence $\{x_l\}_{l=1}^{\infty} \subset K$ such that $f_{k_l}(x_l) \notin U$. By the compactness of K, we can choose a subsequence $\left\{x_{l_j}\right\}_{j=1}^{\infty}$ that converges to a point $x \in K$. Moreover, since $f \in V(K, U)$, we have $f(K) \subset U$, and therefore $f(x) \in U$, i.e. U is a neighborhood of $f(x)$. However, we have $f_{k_{l_j}}\left(x_{l_j}\right) \notin U$ for all $j \in \mathbb{N}$, a contradiction with the condition that $f_k(x_k) \to f(x)$ whenever $x_k \to x$. ∎

Definition 2.2.8. A sequence $\{f_k\}_{k=1}^{\infty}$ of functions between topological spaces X and Y is said to converge *continuously* to $f : X \to Y$ if, whenever $x_k \to x \in X$, we have the convergence $f_k(x_k) \to f(x) \in Y$.

Corollary 2.2.9. *From the above lemma we conclude that if $n \geq 1$, a sequence of functions $\{f_k\}_{k=1}^{\infty}$ in $C_n(\mathbb{R})$ converges to $f \in C_n(\mathbb{R})$ if and only if it converges continuously.*

2.3. Differentiable functions

Finally, we define the spaces of curried differentiable functions.

Definition 2.3.1. Let $m \in \mathbb{N}_0$ and $n \in \mathbb{N}_0$. The set $D_n^m(\mathbb{R})$ of *curried m times differentiable functions* is a topological space defined recursively as follows:

- If $n = 0$, we set $D_0^m(\mathbb{R}) = \mathbb{R}$ for all m, as usual;
- If $m = 0$, we set $D_n^0(\mathbb{R}) = C_n(\mathbb{R})$;
- Let $m, n > 0$, and assume that both $D_{n-1}^m(\mathbb{R})$ and $D_n^{m-1}(\mathbb{R})$ are defined. Then we let $D_n^m(\mathbb{R})$ be the set of all functions $f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ such that there is a function $f' \in D_n^{m-1}(\mathbb{R})$ which satisfies

$$\frac{f(x+h)-f(x)}{h} \xrightarrow[h\to 0]{} f'(x) \in H_{n-1}(\mathbb{R})$$

for all $x \in \mathbb{R}$, where **the convergence is taken in the space $\boldsymbol{H_{n-1}}(\mathbb{R})$**. Clearly, if $f, g \in D_n^m(\mathbb{R})$ and $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha f + \beta g)' = \alpha f' + \beta g',$$

so in particular, $\alpha f + \beta g \in D_n^m(\mathbb{R})$.

We endow $D_n^m(\mathbb{R})$ with the coarsest topology in which the maps

$$\begin{aligned} i : D_n^m(\mathbb{R}) &\to \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})), & \qquad d : D_n^m(\mathbb{R}) &\to D_n^{m-1}(\mathbb{R}) \\ f &\mapsto f & f &\mapsto f' \end{aligned} \tag{1}$$

are continuous. This topology will be called the *differential topology of order m*. Corollary 2.3.4 and Lemma B.3 ensure that $D_n^m(\mathbb{R})$ is a Hausdorff topological space.

Lemma 2.3.2. *For all $m, n \in \mathbb{N}_0$, we have $D_n^{m+1}(\mathbb{R}) \subset D_n^m$, and the inclusion map*

$$j : D_n^{m+1}(\mathbb{R}) \to D_n^m(\mathbb{R})$$

is continuous, meaning that the topology on $D_n^{m+1}(\mathbb{R})$ is finer than the topology induced from $D_n^m(\mathbb{R})$.

Proof: We prove by induction over n. Consider the following cases:

1. $n = 0$. Then we have $D_0^{m+1}(\mathbb{R}) = D_0^n(\mathbb{R}) = \mathbb{R}$, and the statement holds;
2. $m = 0$. Then we have

$$D_n^1(\mathbb{R}) \subset \mathcal{C}(\mathbb{R}, D_{n-1}^0(\mathbb{R})) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R})) = C_n(\mathbb{R}) = D_n^0(\mathbb{R}),$$

   and the inclusion is continuous by the definition of differential topology.
3. $n, m > 0$, with the statement proven for $(m, n-1)$ and $(m-1, n)$. Let $f \in D_n^{m+1}(\mathbb{R})$. Then we have

$$f \in \mathcal{C}\big(\mathbb{R}, D_{n-1}^{m+1}(\mathbb{R})\big) \subset \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})).$$

   Moreover, since $f' \in D_n^m(\mathbb{R}) \subset D_n^{m-1}(\mathbb{R})$, we see that $f \in D_n^m(\mathbb{R})$ by definition, so we have the inclusion $D_n^{m+1}(\mathbb{R}) \subset D_n^m(\mathbb{R})$. It only remains to show that the inclusion map $j : D_n^{m+1}(\mathbb{R}) \to D_n^m(\mathbb{R})$ is continuous. This is equivalent to showing that the maps $i \circ j$ and $d \circ j$ are continuous, which is evident from the following commutative diagrams:

$$\begin{tikzcd} D_n^{m+1}(\mathbb{R}) \arrow[r, hook, "j"] \arrow[d, hook, "i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\ \mathcal{C}\big(\mathbb{R}, D_{n-1}^{m+1}(\mathbb{R})\big) \arrow[r, hook, "j"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \end{tikzcd} \qquad \begin{tikzcd} D_n^{m+1}(\mathbb{R}) \arrow[r, hook, "j"] \arrow[d, "d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\ D_n^m(\mathbb{R}) \arrow[r, hook, "j"'] & D_n^{m-1}(\mathbb{R}) \end{tikzcd}$$

■

Corollary 2.3.3. *We have a chain of subspaces with each topology finer than the previous one:*

$$C_n(\mathbb{R}) = D_n^0(\mathbb{R}) \xleftarrow{j} D_n^1(\mathbb{R}) \xleftarrow{j} D_n^2(\mathbb{R}) \xleftarrow{j} D_n^3(\mathbb{R}) \xleftarrow{j} \dots$$

Corollary 2.3.4. *For all $n, m \in \mathbb{N}_0$, the topological space $D_n^m(\mathbb{R})$ is Hausdorff.*

Proof: Since $C_n(\mathbb{R})$ is Hausdorff and $i : D_n^m(\mathbb{R}) \to C_n(\mathbb{R})$ is continuous and injective, we see that $D_n(\mathbb{R})$ is also Hausdorff. ■

Definition 2.3.5. Let $n \in \mathbb{N}_0$. We define the space $D_n^\infty(\mathbb{R})$ of *curried infinitely differentiable* functions as follows:

$$D_n^\infty(\mathbb{R}) = \bigcap_{m=0}^{\infty} D_n^m(\mathbb{R}).$$

The topology on $D_n^\infty(\mathbb{R})$ is the initial topology with respect to the family of inclusion maps $\{i_m\}_{m=0}^\infty$, where $i_m : D_n^\infty \to D_n^m(\mathbb{R})$.

Lemma 2.3.6. *For every $n \in \mathbb{N}_0$, the space D_n^∞ is a Fréchet space.*

Proof: A trivial exercise. ■

3. Correspondence with uncurried functions

Any function from $\mathbb{R}^n \to \mathbb{R}$ can be rewritten as an element of $H_n(\mathbb{R})$, and vice versa. Naturally, one might ask whether the spaces $C_n(\mathbb{R})$ and $D_n^m(\mathbb{R})$ correspond similarly to continuous and continuously differentiable, respectively, functions from $\mathbb{R}^n$ to $\mathbb{R}$. The answer to this question is yes.

Definition 3.1. Let $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ be a function. Define

$$\gamma\left(\tilde{f}\right)(x_1)(x_2)...(x_n) = \tilde{f}(x_1, x_2, ..., x_n).$$

Then $\gamma : \operatorname{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ is clearly a bijection. The function $f = \gamma\left(\tilde{f}\right)$ will be called the *curry* of $\tilde{f}$.

Lemma 3.2. *Let $n \in \mathbb{N}$. If $\operatorname{Hom}(\mathbb{R}^n, \mathbb{R})$ is given the topology of pointwise convergence together with the pointwise linear structure, then $\gamma : \operatorname{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ becomes a linear homeomorphism of topological vector spaces.*

Proof: We easily see by induction that a filter $F \in \mathcal{F}(H_n(\mathbb{R}))$ converges to $f \in H_n(\mathbb{R})$ if and only if $F(x_1)(x_2)...(x_n) \to f(x_1)(x_2)...(x_n) \in \mathbb{R}$ for any points $x_1, ..., x_n \in \mathbb{R}$. But this is equivalent to $\gamma^{-1}(F) \to \gamma^{-1}(f)$ in $\operatorname{Hom}(\mathbb{R}^n, \mathbb{R})$, so we see that γ is a homeomorphism. ■

Now, let us consider the spaces $C_n(\mathbb{R})$ of continuous functions.

Proposition 3.3. (see[1], page 302) *Let X, Y, Z be topological spaces such that X is Hausdorff and Y is locally compact. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y, Z))$$

defined as $\gamma(f)(x)(y) = f(x, y)$, is well-defined and is a homeomorphism.

Corollary 3.4. *Let $n \in \mathbb{N}_0$. Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \to C_n(\mathbb{R})$$

defined by $\gamma\left(\tilde{f}\right)(x_1)(x_2)...(x_n) = \tilde{f}(x_1, ..., x_n)$, is well-defined and is a homeomorphism.

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By Proposition 3.3, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ■

A similar result holds for the spaces $D^m_n(\mathbb{R})$ which are in a natural isomorphism with the spaces $C^m(\mathbb{R}^n)$.

Definition 3.5. Consider the space $C^m(\mathbb{R}^n)$ of functions $f : \mathbb{R}^n \to \mathbb{R}$ that have continuous partial derivatives of order up to and including m. For $\alpha = (i_1, i_2, ..., i_s)$, where $s \leq m$ and $1 \leq i_k \leq n$ for $k \in \{1, ..., s\}$, write $|\alpha| = s$ and

$$D^\alpha f = \frac{\partial^s f}{\partial x_{i_1} \partial x_{i_2} ... \partial x_{i_s}}.$$

Now, let K be a compact subset of $\mathbb{R}^n$. Define

$$\|f\|_K = \max_{|\alpha| \leq m} \max_{x \in K} |D^\alpha f(x)|.$$

Choosing a sequence $\{K_j\}_{j=1}^\infty$ such that $\mathbb{R}^n = \bigcup_{j=1}^\infty K_j$, we obtain a separating family $\left\{\|\cdot\|_{K_j}\right\}_{j=1}^\infty$ of seminorms, which generates a metrizable vector topology on $C^m(\mathbb{R}^n)$. Note that if $m = 0$, we obtain the familiar compact-open topology on $\mathcal{C}(\mathbb{R}^n, \mathbb{R})$.

Theorem 3.6. *Let $m, n \in \mathbb{N}$. Then the curry map*

$$\gamma : C^m(\mathbb{R}^n) \leftrightarrow D^m_n(\mathbb{R})$$

is well-defined and is an isomorphism of topological vector spaces. Moreover, for each $1 \leq i_1 \leq i_2 \leq ... \leq i_m \leq n$, we have

$$\frac{\partial^m \tilde{f}}{\partial x_{i_1} \partial x_{i_2} ... \partial x_{i_m}}(x_1, x_2, ..., x_n) = f(x_1)...\left(x_{i_1-1}\right)'\left(x_{i_1}\right)...\left(x_{i_2-1}\right)'\left(x_{i_2}\right)...\left(x_{i_m-1}\right)'\left(x_{i_m}\right)...(x_n). \quad (2)$$

Proof: If $n = 0$ or $m = 0$, the statement is trivial. Now let $n, m > 0$ and assume the statement proven for $(m, n-1)$ and $(m-1, n)$.

First, let $f \in D^m_n(\mathbb{R})$. We immediately see that $f \in D^{m-1}_n(\mathbb{R})$, and so, by the induction hypothesis, partial derivatives or order up to $m-1$ exist and are continuous on $\mathbb{R}^n$. Now, consider some $1 \leq i_1 \leq i_2 \leq ..., \leq i_m \leq n$. It is not hard to see by definition that (2) holds. To show that the partial derivative with respect to $x_{i_1}, ..., x_{i_m}$ is continuous, consider $(h_1, h_2, ..., h_n) \to 0 \in \mathbb{R}^n$, and a point $(x_1, ..., x_n) \in \mathbb{R}^n$. By the continuity of f and the definition of the differential topology, we have

$$\begin{aligned} f(x_1 + h_1) &\xrightarrow[h_1 \to 0]{} f(x_1) \in D^m_{n-1}(\mathbb{R}), \\ &\vdots \\ f(x_1 + h_1)...\left(x_{i_1-1} + h_{i_1-1}\right)' &\xrightarrow[h_1, ..., h_{i_1-1} \to 0]{} f(x_1)...\left(x_{i_1-1}\right)' \in D^{m-1}_{n-i_1+1}(\mathbb{R}), \\ &\vdots \\ &f(x_1 + h_1)...\left(x_{i_1-1} + h_{i_1-1}\right)'...\left(x_{i_m-1} + h_{i_m-1}\right)...(x_n + h_n) \\ &\xrightarrow[h_1, h_2, ..., h_n \to 0]{} f(x_1)...\left(x_{i_1-1}\right)'...\left(x_{i_m-1}\right)'...(x_n) \in D^0_0(\mathbb{R}) = \mathbb{R}, \end{aligned}$$

so the partial derivative

$$\frac{\partial^m \tilde{f}}{\partial x_{i_1} \partial x_{i_2} ... \partial x_{i_m}}$$

is continuous on $\mathbb{R}^n$. Note that so far we have only proven the existence and continuity of partial derivatives with non-decreasing indices $i_1, ..., i_m$. However, as stated in[3] (pp. 235-236), the order of partial differentiation does not matter, as long as one of the partial derivatives is shown to be continuous.

Conversely, suppose $f \in H_n(\mathbb{R})$ is such that $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ lies in the class $C^m(\mathbb{R}^n)$. To show that $f \in D_n^m(\mathbb{R})$, we follow three steps:

1. For all $x_1 \in \mathbb{R}$, we have $f(x_1) \in D_{n-1}^m(\mathbb{R})$. This is provided by the induction hypothesis, since the restriction of a C^m function on a hyperplane is also a C^m function.
2. The function $f : \mathbb{R} \to D_{n-1}^m(\mathbb{R})$ is continuous. To show this, consider a sequence $\left\{x_k^{(1)}\right\}_{k\in\mathbb{N}} \subset \mathbb{R}$ converging to $x_0^{(1)} \in \mathbb{R}$. We aim to prove that $f\left(x_k^{(1)}\right) \to f\left(x_0^{(1)}\right)$ in $D_{n-1}^m(\mathbb{R})$. This is equivalent to showing that

$$f\left(x_k^{(1)}\right) \to f\left(x_k^{(1)}\right) \in \mathcal{C}(\mathbb{R}, D_{n-2}^m(\mathbb{R})) \quad \text{and} \quad f\left(x_k^{(1)}\right)' \to f\left(x_0^{(1)}\right)' \in D_{n-1}^{m-1}(\mathbb{R}). \tag{3}$$

   The latter of these conditions holds by the induction hypothesis, since

$$\frac{\partial \tilde{f}}{\partial x_2}(x_1, ..., x_n) = \frac{\partial \widetilde{f(x_1)}}{\partial x_2}(x_2, ..., x_n) \underset{\text{by (2)}}{=} f(x_1)'(x_2)...(x_n), \tag{4}$$

   and $\partial \tilde{f} / \partial x_2$ is continuous, which means that the map $x \mapsto f(x)'$ is continuous. To prove the former condition in (3), we again consider a sequence $x_k^{(2)} \to x_0^{(2)} \in \mathbb{R}$ and aim to prove

$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right) \in D_{n-2}(\mathbb{R}),$$

   which (unless $n = 2$) is again equivalent to

$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \quad \text{and} \quad f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)' \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)'.$$

   The latter follows from the induction hypothesis and the continuity of $\partial \tilde{f} / \partial x_3$, similarly to (4), and the former can again be expanded further. This chain leads us finally to the convergence

$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)...\left(x_k^{(n)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)...\left(x_0^{(n)}\right) \in \mathbb{R},$$

   which is evident since $f \in C_n(\mathbb{R})$ by Corollary 3.4. Hence we see that f lies in $\mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R}))$. We have also established (2) for f and $i \geq 2$, via (4).
3. Finally, we need to find $f' \in D_n^{m-1}(\mathbb{R})$ such that

$$\frac{f(x+h) - f(x)}{h} \underset{h\to 0}{\longrightarrow} f'(x) \in H_{n-1}(\mathbb{R}) \tag{5}$$

   for all $x \in \mathbb{R}$. Indeed, following (2), let us define $f' := \gamma\left(\partial \tilde{f} / \partial x_1\right)$. We immediately see by the induction hypothesis that f' so defined lies in $D_n^{m-1}(\mathbb{R})$. To see that (5) holds, consider $x_1, ..., x_n \in \mathbb{R}$ and write

$$\begin{aligned} f'(x_1)(x_2)...(x_n) &= \frac{\partial \tilde{f}}{\partial x_1}(x_1, ..., x_n) \\ &= \lim_{h \to 0} \frac{\tilde{f}(x_1 + h, x_2, ..., x_n) - \tilde{f}(x_1, ..., x_n)}{h} \\ &= \left(\lim_{h \to 0} \frac{f(x_1 + h) - f(x_1)}{h} \right)(x_2)...(x_n), \end{aligned}$$

and we are done.

Thus we have shown that $\gamma : C^m(\mathbb{R}^n) \to D^m_n(\mathbb{R})$ is a bijection. It is clearly a linear map, so it only remains to show that γ is continuous. Let $\tilde{f}_k \in C^m(\mathbb{R}^n)$ with $\tilde{f}_k \xrightarrow[k\to\infty]{} \tilde{f} \in C^m(\mathbb{R}^n)$. We aim to show that $f_k = \gamma\left(\tilde{f}_k\right) \xrightarrow[k\to\infty]{} \gamma\left(\tilde{f}\right) = f \in D^m_n(\mathbb{R})$. Indeed, this is equivalent to showing that

$$f_k \xrightarrow[k\to\infty]{} f \in \mathcal{C}(\mathbb{R}, D^m_{n-1}(\mathbb{R})) \quad \text{and} \quad f'_k \xrightarrow[k\to\infty]{} f' \in D^{m-1}_n(\mathbb{R}).$$

For the former condition, consider a sequence $x_k \to x \in \mathbb{R}$, and note that $\gamma^{-1}(f_k(x_k)) \in C^m\left(\mathbb{R}^{n-1}\right)$, so by the induction hypothesis we have

$$f_k(x_k) \xrightarrow[k\to\infty]{} f(x) \in D^m_{n-1}(\mathbb{R}).$$

For the latter condition, note that

$$\gamma^{-1}(f'_k) = \frac{\partial \tilde{f}}{\partial x_1} \in C^{m-1}(\mathbb{R}^n),$$

hence, applying the induction hypothesis again, we have

$$f'_k \xrightarrow[k\to\infty]{} f \in D^{m-1}_n(\mathbb{R}),$$

and the proof of the theorem is concluded. ■

4. Completeness and metrizability

In this section, we will establish some strong properties of the spaces $C_n(\mathbb{R})$ and $D^m_n(\mathbb{R})$, namely local convexity, metrizability and completeness.

Definition 4.1. A topological vector space X is called *metrizable* if its topology is compatible with an invariant metric d, i.e. such that $d(x + y, x + z) = d(y, z)$ for all $x, y, z \in X$.

Definition 4.2. Let X be a topological vector space, and let $\{x_k\}_{k=1}^{\infty} \subset X$ be a sequence. This sequence is called a *Cauchy sequence* if, for any neighborhood U of zero, there is $N \in \mathbb{N}$ such that for all $k, l \geq N$, we have $x_k - x_l \in U$. Clearly, if X is metrizable, this definition corresponds to the metric definition of a Cauchy sequence.
If every Cauchy sequence in X has a limit in X, we call X a *sequentially complete* topological vector space.

Definition 4.3. A topological vector space is called a *Fréchet space* if it is locally convex, metrizable, and sequentially complete. In other words, if its topology is compatible with a complete invariant metric.

A very convenient criterion of metrizability is the existence of a countable local basis:

Proposition 4.4. (see[4], page 18) *A topological vector space X is metrizable if and only if it admits a countable local basis of neighborhoods of* 0.

Now, let us first consider the general spaces $H_n(\mathbb{R})$.

Theorem 4.5. *For all $n \in \mathbb{N}_0$, the space H_n is locally convex and sequentially complete, but not metrizable for $n > 0$.*

Proof: If $n = 0$, we have $H_0(\mathbb{R}) = \mathbb{R}$, which is clearly locally convex, sequentially complete and metrizable.

Now, let $n > 0$ and assume that $H_{n-1}(\mathbb{R})$ is locally convex and sequentially complete. Consider a Cauchy sequence $\{f_k\}_{k=1}^{\infty} \subset H_n(\mathbb{R})$. It is easy to see by the definition of the pointwise convergence topology, that for all $x \in \mathbb{R}$, the sequence $\{f_k(x)\}_{k=1}^{\infty} \subset H_{n-1}(\mathbb{R})$ is Cauchy, hence it has a limit which we denote by $f(x)$. We have therefore constructed a function $f \in H_n(\mathbb{R})$ with $f_k \xrightarrow[k\to\infty]{} f$, and we can conclude that $H_n(\mathbb{R})$ is complete.

Next, we show that $H_n(\mathbb{R})$ is locally convex. It suffices to prove that any neighborhood of the form $V(x, U)$ contains a convex neighborhood V' of zero. Indeed, since $H_{n-1}(\mathbb{R})$ is locally convex, let $U' \subset U$ be a convex neighborhood of zero in $H_{n-1}(\mathbb{R})$. Then we have $0 \in V(x, U') \subset V(x, U)$, and $V(x, U')$ is clearly convex.

Finally, to show that $H_n(\mathbb{R})$ is **not** metrizable, it suffices to show that no countable local basis exists in $H_n(\mathbb{R})$. To this end, assume the contrary, i.e. let $\{V_j\}_{j=1}^{\infty}$ be a countable local basis in $H_n(\mathbb{R})$. Since every set V_j contains a neighborhood of zero of the form

$$\bigcap_{l=1}^{N_j} V\left(x_l^j, U_l^j\right),$$

where $x_l^j \in \mathbb{R}$ and U_l^j are neighborhoods of zero in $H_{n-1}(\mathbb{R})$, by uniting all these $V\left(x_l^j, U_l^j\right)$ into one family we obtain a countable local subbase $\{V(x_i, U_i)\}_{i=1}^{\infty}$ in $H_n(\mathbb{R})$. In particular, it follows that a sequence of functions $\{f_k\}_{k=1}^{\infty} \subset H_n(\mathbb{R})$ converges to 0 if and only if for all $i \in \mathbb{N}$, eventually $f_k \in V(x_i, U_i)$. However, we can easily choose $\varphi \in H_{n-1}(\mathbb{R})$ such that $\varphi \neq 0$, and then define

$$f_k(x) = \begin{cases} 0, & \text{if } x = x_i \text{ for some } i, \\ \varphi, & \text{otherwise.} \end{cases}$$

For all $i, k \in \mathbb{N}$, we have $f_k(x_i) = 0 \in U_i$, so $f_k \in V(x_i, U_i)$. However, clearly $\{f_k\}_{k=1}^{\infty}$ does not converge to 0. A contradiction. ∎

Next, we consider the spaces $C_n(\mathbb{R})$ of continuous functions, which turn out to be both complete and metrizable.

Theorem 4.6. *Suppose that X is a locally compact and hemicompact topological space (that is, X is the union of countably many its compact subsets), and let Y be a Fréchet space. Then the space $\mathcal{C}(X, Y)$ is also a Fréchet space.*

Proof: First of all, we need to show that $\mathcal{C}(X, Y)$ is metrizable. Since Y is metrizable, by Proposition 4.4 it admits a countable basis $\{U_n\}_{n=1}^{\infty}$ of neighborhoods of 0. Moreover, since X is hemicompact, we have

$$X = \bigcup_{m=1}^{\infty} K_m,$$

where all K_m are compact. It is clear to see that the family

$$\{V(K_m, U_n) \mid m, n \in \mathbb{N}\}$$

forms a countable basis of neighborhoods of 0 in $\mathcal{C}(X, Y)$. Hence, by Proposition 4.4 the space $\mathcal{C}(X, Y)$ is metrizable.

Next, we prove that $\mathcal{C}(X, Y)$ is locally convex. Let $m, n \in \mathbb{N}$. We need to find a convex neighborhood of zero inside $V(K_m, U_n)$. Since Y is locally convex, we have a convex neighborhood $U' \subset U_n$ of zero. Now simply take $V' = V(K_m, U')$, and observe that $V' \subset V(K_m, U_n)$ and V' is convex.

It remains to show that $\mathcal{C}(X, Y)$ is complete. To this end, consider a Cauchy sequence $\{f_k\}_{k=1}^{\infty}$ in $\mathcal{C}(X, Y)$. This means that for all $n, m \in \mathbb{N}$, there is $N \in \mathbb{N}$ such that

$$k, l \geq N \Longrightarrow f_k - f_l \in V(K_m, U_n).$$

Now, consider a point $x \in X$. By choosing m_0 such that $x \in K_{m_0}$, we see that for all $n \in \mathbb{N}$, eventually

$$f_k - f_l \in V\left(K_{m_0}, U_n\right) \Longrightarrow f_k(x) - f_l(x) \in U_n.$$

This means that the sequence $\{f_k(x)\}_{k=1}^{\infty}$ is Cauchy in Y, and so it converges to some point $f(x) \in Y$. We now need to show that f is continuous and $f_k \to f$ in $\mathcal{C}(X, Y)$.
Consider $n, m \in \mathbb{N}$. Since Y is a topological vector space, it is *regular* in the sense that every neighborhood U of 0 contains the closure $\overline{U'}$ of some other neighborhood U' of 0. Hence, we can find $n' \in \mathbb{N}$ such that $\overline{U_{n'}} \subset U_n$. Now, as $k, l \to \infty$, eventually we have

$$(f_k - f_l)(K_m) \subset U_{n'}.$$

Taking any $x \in K_m$, we have

$$f_k(x) - f_l(x) = (f_k - f_l)(x) \in U_{n'} \underset{l \to \infty}{\Longrightarrow} f_k(x) - f(x) \in \overline{U_{n'}} \subset U_n.$$

This means that

$$(f_k - f)(K_m) \subset U_n \tag{6}$$

eventually as $k \to \infty$. Now, to show that f is continuous, consider $x \in X$ and a neighborhood U of $0 \in Y$. Since X is locally compact, there is m_0 such that K_{m_0} is a neighborhood of x. Moreover, there is $n_0 \in \mathbb{N}$ such that $U_{n_0} + U_{n_0} + U_{n_0} \subset U$. Finally, by (6) there is $k_0 \in \mathbb{N}$ such that $\left(f_{k_0} - f\right)\left(K_{m_0}\right) \subset U_{n_0}$. Now, as $y \to x$, we have $y \in K_{m_0}$, and

$$f(y) - f(x) = [f(y) - f_k(y)] + [f_k(y) - f_k(x)] + [f_k(x) - f(x)] \in U_{n_0} + U_{n_0} + U_{n_0} \subset U.$$

Hence, we conclude that f is continuous, and (6) implies that for all $m, n \in \mathbb{N}$, we have $f_k - f \in V(K_m, U_n)$ eventually as $k \to \infty$, which means that $f_k \to f$ in $C(X, Y)$, as desired. ■

Corollary 4.7. *For all $n \in \mathbb{N}_0$, the space $C_n(\mathbb{R})$ is a Fréchet space.*

Proof: If $n = 0$, the statement clearly holds. Now, assuming that $C_{n-1}(\mathbb{R})$ is a Fréchet space, we easily conclude by Theorem 4.6 that $C_n(\mathbb{R})$ is also a Fréchet space. ■

Finally, we examine the spaces $D_n^m(\mathbb{R})$ of differentiable functions. To show that these spaces are Fréchet, we need some preliminary results.

Definition 4.8. A sequence of functions $f_k : X \to Y$ between topological vector spaces X and Y is said to converge *uniformly* to a function $f : X \to Y$ on a subset $E \subset X$ if, for any neighborhood U of $0 \in Y$, we have $f_k(E) \subset U$ eventually as $k \to \infty$.

Lemma 4.9. *Let X and Y be topological vector spaces and let $f_k \to f$ in the space $\mathcal{C}(X,Y)$. Then f_k converges uniformly to f on all compact subsets of X.*

Proof: Let $K \subset X$ be compact and let U be a neighborhood of $0 \in Y$. Since $f_k \to f \in \mathcal{C}(X,Y)$, we have $f_k - f \in V(K,U)$ eventually as $k \to \infty$. This is to say that eventually $(f_k - f)(K) \subset U$, implying that $\{f_k\}_{k=1}^{\infty}$ converges to f uniformly on K. ■

Theorem 4.10. *For all $m, n \in \mathbb{N}_0$, the space $D_n^m(\mathbb{R})$ is a Fréchet space.*

Proof: If either $m = 0$ or $n = 0$, then the statement is either trivial or proven before. Suppose that $m, n > 0$, and that the spaces $D_n^{m-1}(\mathbb{R})$ and $D_{n-1}^m(\mathbb{R})$ are Fréchet spaces. By Theorem 4.6 we see that the space $\mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ is also Fréchet.
First, we have to show that $D_n^m(\mathbb{R})$ is metrizable. Its topology is generated by sets of the form $i^{-1}(V) \cap d^{-1}(W)$, where V and W are open sets in $\mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ and $D_n^{m-1}(\mathbb{R})$ respectively. By the induction hypothesis, we know that both of these spaces are metrizable, and so have countable local bases:

$$\left\{0 \in V_j \subset \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \,\middle|\, j \in \mathbb{N}\right\}, \qquad \left\{0 \in W_k \subset D_n^{m-1}(\mathbb{R}) \,\middle|\, k \in \mathbb{N}\right\}.$$

We shall show that the family $\left\{i^{-1}(V_j) \cap d^{-1}(W_k)\right\}_{j,k=1}^{\infty}$ forms a local basis in $D_n^m(\mathbb{R})$. Indeed, let U be a neighborhood of zero in $D_n^m(\mathbb{R})$. Then we have

$$U \supset i^{-1}(V) \cap d^{-1}(W) \supset i^{-1}(V_j) \cap d^{-1}(W_k) \ni 0$$

for some open sets $V \subset \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})), W \subset D_n^{m-1}(\mathbb{R})$ and some $j, k \in \mathbb{N}$.

Second, we must show that $D_n^m(\mathbb{R})$ is locally convex. Consider a neighborhood U of $0 \in D_n^m(\mathbb{R})$. We can assume without loss of generality that $U = i^{-1}(V) \cap d^{-1}(W)$, where V and W are neighborhoods of zero in $\mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ and $D_n^{m-1}(\mathbb{R})$, respectively. Since both of these spaces are Fréchet, we have convex neighborhoods $V' \subset V$ and $W' \subset W$. Lastly, since both maps i, d are linear, we see that the set

$$0 \in U' = i^{-1}(V') \cap d^{-1}(W') \subset U$$

is convex. Therefore, the space $D_n^m(\mathbb{R})$ is locally convex.

Third, we need to prove that all Cauchy sequences in $D_n^m(\mathbb{R})$ have a limit. To this end, let $\{f_k\}_{k=1}^{\infty} \subset D_n^m(\mathbb{R})$ be a Cauchy sequence. Then, clearly, the sequences $\{f_k\}_{n=1}^{\infty} \subset \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ and $\{f_k'\}_{k=1}^{\infty} \subset D_n^{m-1}(\mathbb{R})$ are Cauchy, meaning that we have

$$f_k \xrightarrow[k\to\infty]{} f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})), \qquad f_k' \xrightarrow[k\to\infty]{} g \in D_n^{m-1}(\mathbb{R}).$$

We must now show that $f \in D_n^m(\mathbb{R})$ with $f' = g$. To this end, let $x_0^{(2)}, x_0^{(3)}, ..., x_0^{(n)} \in \mathbb{R}$ and define the auxiliary functions

$$\varphi_k(x) = f_k(x)\left(x_0^{(2)}\right)...\left(x_0^{(n)}\right),$$

$$\varphi(x) = f(x)\left(x_0^{(2)}\right)...\left(x_0^{(n)}\right),$$

$$\psi(x) = g(x)\left(x_0^{(2)}\right)...\left(x_0^{(n)}\right).$$

We immediately observe that $\varphi_k \xrightarrow[k\to\infty]{} \varphi$ pointwise on $\mathbb{R}$ Now, let $[a,b] \subset \mathbb{R}$. Since $\tilde{f}_k' \to \tilde{g}$ in the space $\mathcal{C}(\mathbb{R}^n, \mathbb{R})$, we see that $\tilde{f}_k'$ converges to $\tilde{g}$ *uniformly* on the compact set

$$[a,b] \times \left\{x_0^{(2)}\right\} \times \left\{x_0^{(2)}\right\} \times \ldots \times \left\{x_0^{(n)}\right\} \subset \mathbb{R}^n.$$

This implies that we have a uniform convergence $\varphi_k' \to \psi$ on $[a,b]$. Therefore, the function φ is differentiable on (a,b) with $\varphi' = \psi$. Since the interval $[a,b] \subset \mathbb{R}$ was arbitrary, we conclude that $\varphi' = g$ on $\mathbb{R}$. In other words, for all $x_0 \in \mathbb{R}$ we have the pointwise limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = g(x_0).$$

Since $f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ and $g \in D_n^{m-1}(\mathbb{R})$, we conclude that $f \in D_n^m(\mathbb{R})$, and so

$$\left.\begin{array}{l} f_k \to f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \\ f_k' \to f' \in D_n^{m-1}(\mathbb{R}) \end{array}\right\} \Longrightarrow f_k \to f \in D_n^m(\mathbb{R}).$$

Therefore, the space $D_n^m(\mathbb{R})$ is complete, and hence a Fréchet space. ■

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A. The language of filters

Definition A.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition A.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.
If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x.
If $f : X \to Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X, then $\{f(A) \mid A \in F\}$ is a filter basis that generates the *image filter* denoted $f(F)$.

Definition A.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark A.4. The language of filters is expressive enough to encode all topological properties. For example[5]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter

  $$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$

  converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X.

- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

Lemma A.5. (see[5], page 74) *Let X be a set and $f_i : X \to Y_i$ be a family of maps, where Y_i are topological spaces. Let X be endowed with the coarsest topology such that f_i is continuous for each i. Then a filter $F \in \mathcal{F}(X)$ converges to $x \in X$ if and only if $f_i(F)$ converges to $f_i(x)$ for all i.*

B. Continuity of linear operations

Lemma B.1. *Let X be a set and let Y be a real topological vector space. Then $\mathcal{S}(X, Y)$ is also a real topological vector space with respect to the topology of pointwise convergence and the pointwise linear structure.*

Proof: We need to show that the maps $+ : \mathcal{S}(X,Y) \times \mathcal{S}(X,Y) \to \mathcal{S}(X,Y)$ and $\cdot : \mathbb{R} \times \mathcal{S}(X,Y) \to \mathcal{S}(X,Y)$ are continuous.
To begin, consider two filters $F_1, F_2 \in \mathcal{F}(\mathcal{S}(X,Y))$, converging to f_1 and f_2 respectively. To show that $F_1 + F_2$ converges to $f_1 + f_2$, consider $x \in X$. We have

$$\begin{aligned}(F_1 + F_2)(x) &= [\{(A+B)(x) \mid A \in F_1, B \in F_2\}] \\ &= [\{A(x) + B(x) \mid A \in F_1, B \in F_2\}] \\ &= F_1(x) + F_2(x) \to f_1(x) + f_2(x) = (f_1 + f_2)(x).\end{aligned}$$

This shows that the addition operation is continuous. Scalar multiplication is continuous by a similar argument. ∎

Lemma B.2. *Let X be a topological space and Y a real topological vector space. Then $\mathcal{C}(X,Y)$ is also a real topological vector space with respect to the pointwise linear structure and the compact-open topology.*

Proof: We will show that the map $+ : \mathcal{C}(X,Y) \times \mathcal{C}(X,Y) \to \mathcal{C}(X,Y)$ is continuous. Let $F_1, F_2 \in \mathcal{F}(\mathcal{C}(X,Y))$ converge to $f_1 \in \mathcal{C}(X,Y)$ and $f_2 \in \mathcal{C}(X,Y)$ respectively. For any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have

$$\begin{aligned}(F_1 + F_2)(P) &= [\{(A_1 + A_2)(B) \mid A_i \in F_i, B \in P\}] \\ &\supset [\{A_1(B_1) + A_2(B_2) \mid A_i \in F_i, B_i \in P\}] \\ &= F_1(P) + F_2(P) \to f_1(p) + f_2(p) = (f_1 + f_2)(p).\end{aligned}$$

The continuity of scalar multiplication is proven similarly. ∎

Lemma B.3. *For all $m, n \in \mathbb{N}_0$, the pointwise linear structure is compatible with the topology on $D_n^m(\mathbb{R})$, making $D_n^m(\mathbb{R})$ a topological vector space.*

Proof: We prove by induction over n and m. If $n = 0$, we have $D_0(\mathbb{R}) = \mathbb{R}$ with the usual topology and linear structure. If $m = 0$, we have $D_n^0(\mathbb{R}) = C_n(\mathbb{R})$, which is a TVS. Suppose now that $m, n > 0$. We need to establish the continuity of the following two maps:

$$+ : D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \to D_n^m(\mathbb{R}), \qquad \cdot : \mathbb{R} \times D_n^m(\mathbb{R}) \to D_n^m(\mathbb{R}).$$

In turn, by the definition of the differential topology, this task is equivalent to proving that the maps $i \circ (+), d \circ (+), i \circ (\cdot), d \circ (\cdot)$ are continuous, where the maps i and d are defined in (1).
To this end, note that we have commutative diagrams

$$\begin{tikzcd}
D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, hook, "i \times i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\
\mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \times \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \arrow[r, "(+)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))
\end{tikzcd}$$

$$\begin{tikzcd}
D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, "d \times d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\
D_n^{m-1}(\mathbb{R}) \times D_n^{m-1}(\mathbb{R}) \arrow[r, "(+)"'] & D_n^{m-1}(\mathbb{R})
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{R} \times D_n^m(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, hook, "\mathrm{id} \times i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\
\mathbb{R} \times \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \arrow[r, "(\cdot)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{R} \times D_n^m(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, "\mathrm{id} \times d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\
\mathbb{R} \times D_n^{m-1}(\mathbb{R}) \arrow[r, "(\cdot)"'] & D_n^{m-1}(\mathbb{R})
\end{tikzcd}$$

Since the down-then-right path in each diagram is continuous, the right-then-down paths are continuous as well, and we are done. ∎